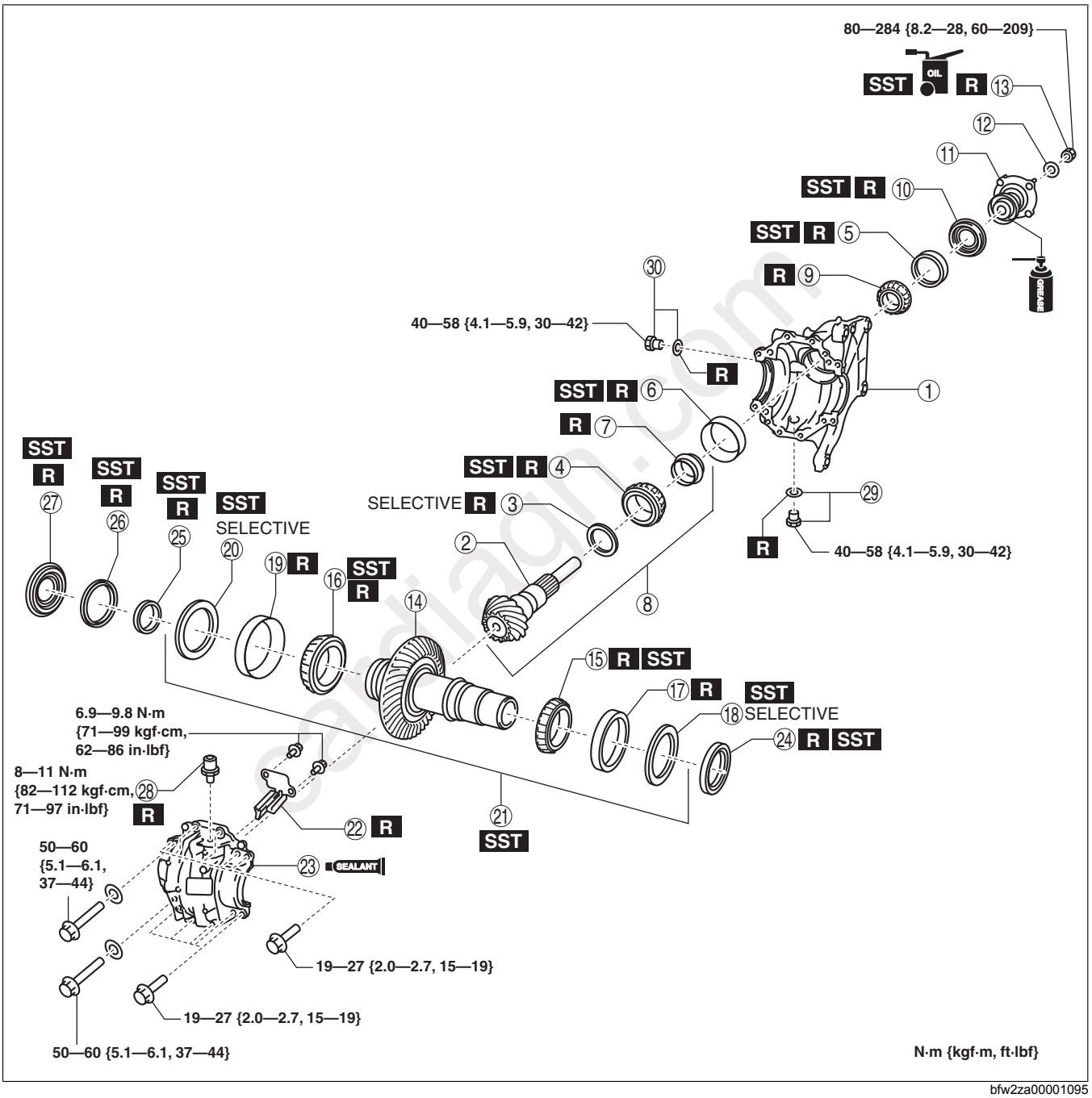
TRANSFER ASSEMBLY [FW6AX-EL]

id0316g1000300

Caution

- **To prevent the silicone sealant from hardening, adhere the drive gear case to the front carrier within 3 min after applying silicone sealant.**
- **Wait 30 min or more after the silicone sealant application area has bonded and fill with transfer oil.**

With tapered roller bearing



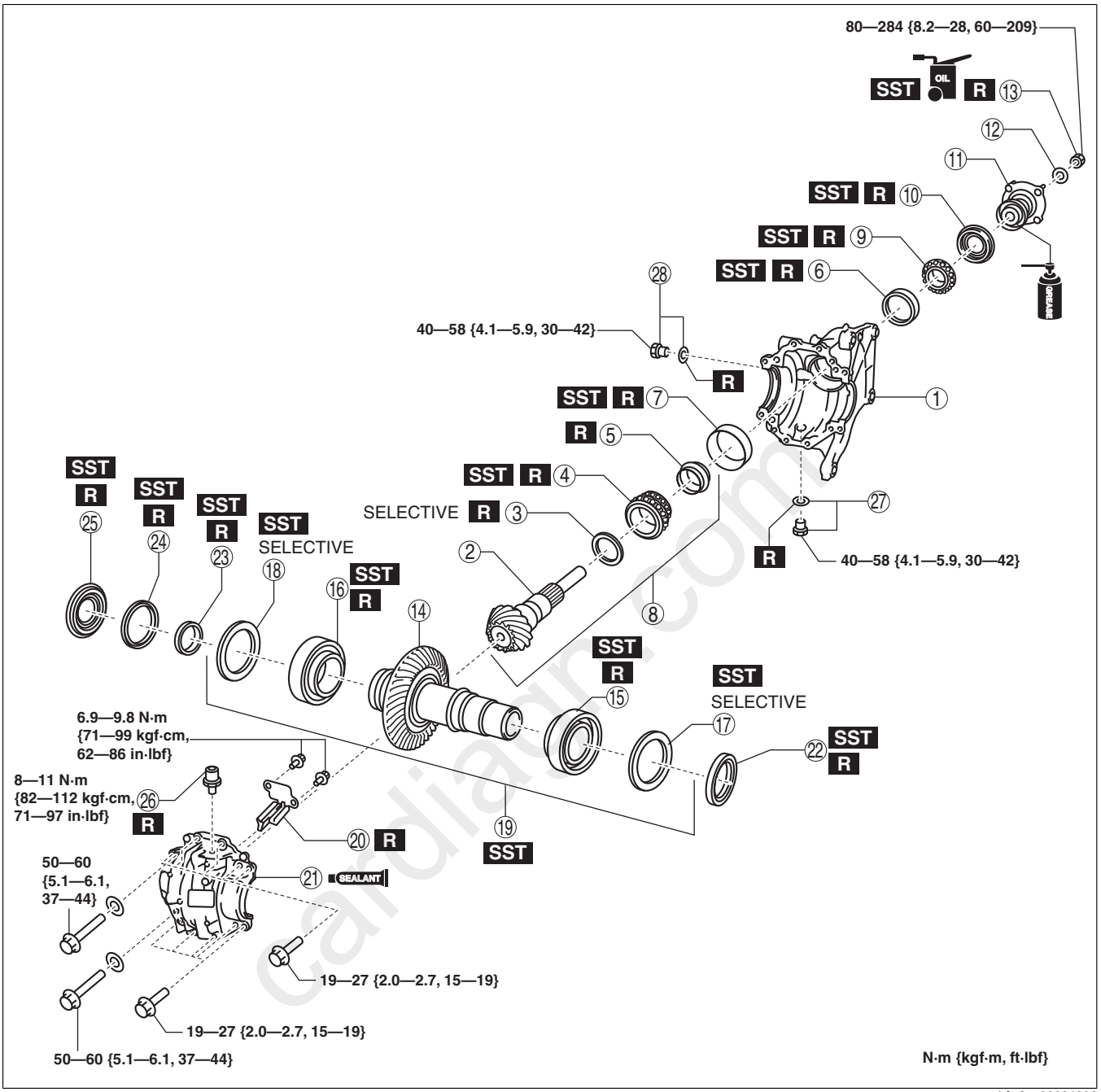
bfw2za00001095

1	Front carrier
2	Drive pinion gear
3	Spacer
4	Bearing inner race (front)
5	Bearing outer race (rear)
6	Bearing outer race (front)
7	Collapsible spacer
8	Drive pinion gear component
9	Bearing inner race (rear)

10	Oil seal
11	Companion flange component
12	Washer
13	Locknut
14	Ring gear shaft
15	Bearing inner race (LH)
16	Bearing inner race (RH)
17	Bearing outer race (LH)
18	Adjustment shim (LH)
19	Bearing outer race (RH)
20	Adjustment shim (RH)
21	Ring gear shaft component
22	Baffle plate
23	Drive gear case
24	Transfer oil seal (LH)
25	Transfer oil seal (RH) No.3
26	Transfer oil seal (RH) No.2
27	Transfer oil seal (RH) No.1
28	Breather
29	Drain plug, washer
30	Oil level plug, washer

cardiagn.com

With ball bearings/tandem ball bearings



bfw2za00001096

1	Front carrier
2	Drive pinion gear
3	Spacer
4	Bearing inner race (front)
5	Collapsible spacer
6	Bearing outer race (rear)
7	Bearing outer race (front)
8	Drive pinion gear component
9	Bearing inner race (rear)
10	Oil seal
11	Companion flange component
12	Washer
13	Locknut
14	Ring gear shaft
15	Ball bearing (LH)
16	Ball bearing (RH)

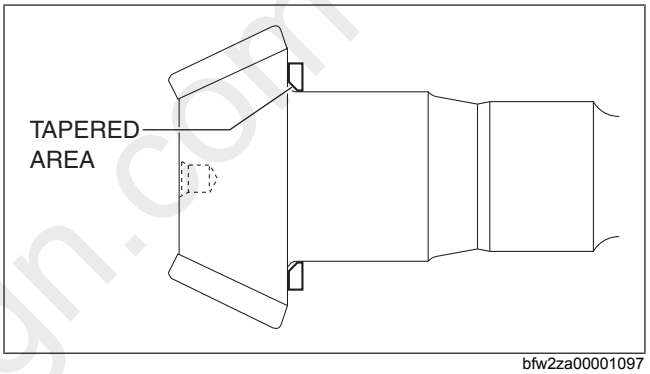
17	Adjustment shim (LH)
18	Adjustment shim (RH)
19	Ring gear shaft component
20	Baffle plate
21	Drive gear case
22	Transfer oil seal (LH)
23	Transfer oil seal (RH) No.3
24	Transfer oil seal (RH) No.2
25	Transfer oil seal (RH) No.1
26	Breather
27	Drain plug, washer
28	Oil level plug, washer

1. Clean the contact surfaces of the front carrier and drive gear case.

Note
- Perform drive pinion preload adjustment with the oil seal not assembled.

2. Assemble the drive pinion gear component and companion flange to the front carrier using the following procedure.
   (1) When disassembling, measure the thickness of the removed spacer and assemble a new spacer with the same dimensions to the drive pinion gear.

Note
- Verify the assembly direction of the spacer and assemble it.



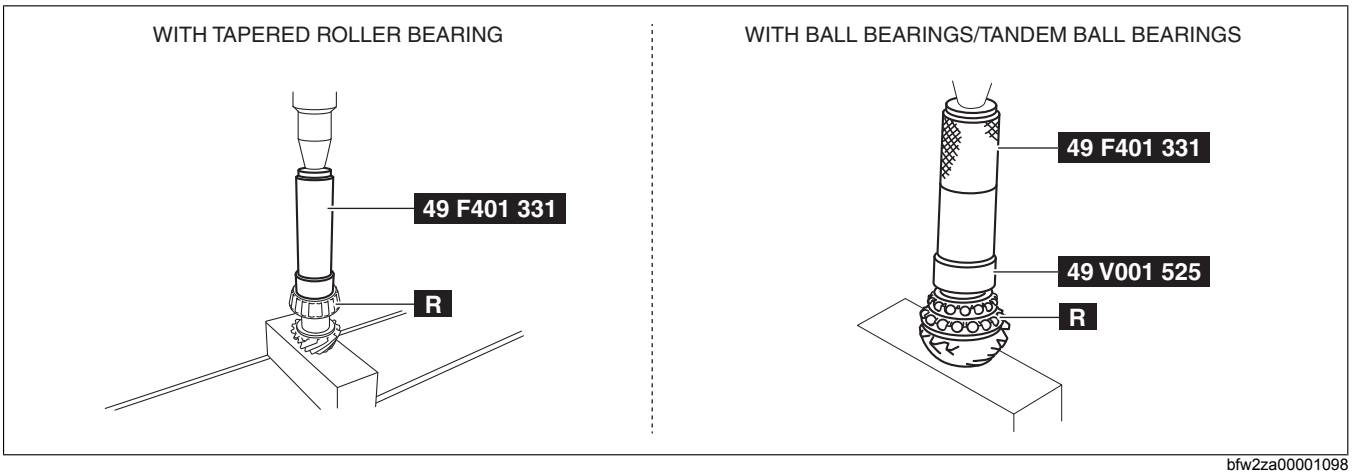
bfw2za00001097

Spacer table

Identification number	Thickness (mm {in})	Identification number	Thickness (mm {in})
08	3.080 {0.1213}	29	3.290 {0.1295}
09	3.095 {0.1219}	30	3.305 {0.1301}
11	3.110 {0.1224}	32	3.320 {0.1307}
12	3.125 {0.1230}	33	3.335 {0.1313}
14	3.140 {0.1236}	35	3.350 {0.1319}
15	3.155 {0.1242}	36	3.365 {0.1325}
17	3.170 {0.1248}	38	3.380 {0.1331}
18	3.185 {0.1254}	39	3.395 {0.1337}
20	3.200 {0.1260}	41	3.410 {0.1343}
21	3.215 {0.1266}	42	3.425 {0.1348}
23	3.230 {0.1272}	44	3.440 {0.1354}
24	3.245 {0.1278}	45	3.455 {0.1360}
26	3.260 {0.1283}	47	3.470 {0.1366}
27	3.275 {0.1289}	–	–

   (2) Press fit the bearing inner race (front) to the drive pinion gear using the **SST** and a press.

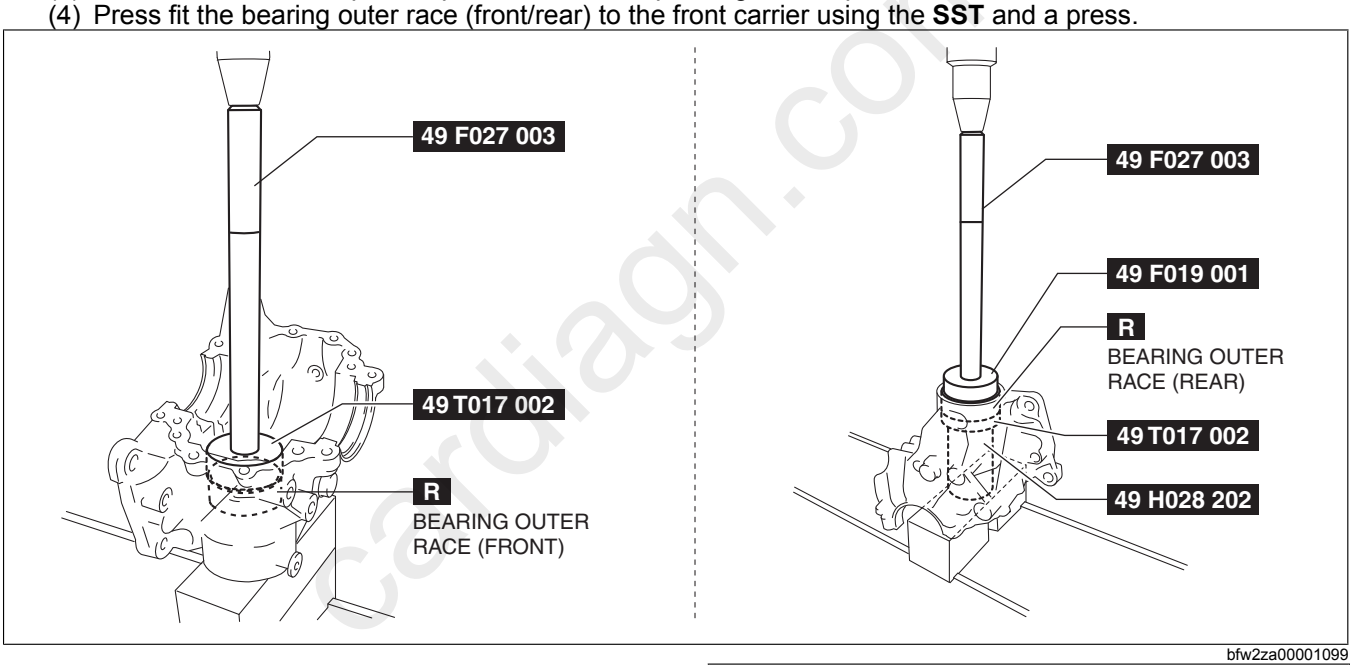
carjiagn.com



bfw2za00001098

Caution
- **Do not apply pressure to the drive pinion teeth with the press. Otherwise, the drive pinion could be damaged.**

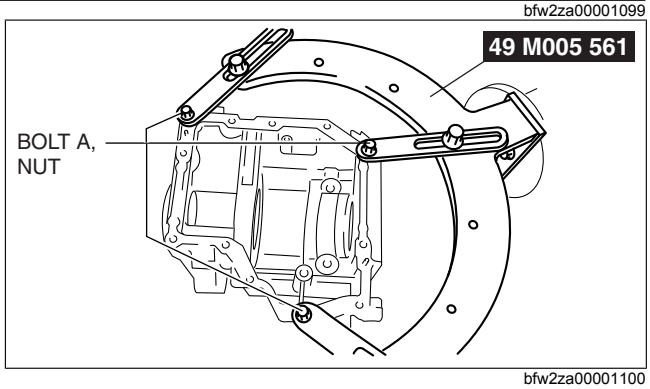
(3) Assemble a new collapsible spacer to the drive pinion gear component.
(4) Press fit the bearing outer race (front/rear) to the front carrier using the **SST** and a press.



bfw2za00001099

(5) Install the front carrier to the **SST**.

Bolts A, nuts
Part number: 9YA02 A242A or M12 x 1.25 bolt (length 42 mm {1.7 in}), and nut (M12 x 1.25)
64—89 N·m {6.6—9.0 kgf·m, 48—65 ft·lbf}



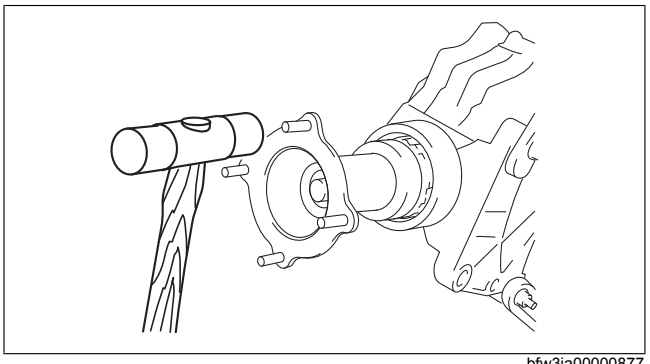
bfw2za00001100

(6) Assemble the drive pinion gear component to the front carrier.
(7) Assemble the bearing inner race (rear) to the drive pinion gear component.
(8) Apply grease to the area of the companion flange where it contacts the bearing inner race (rear).
(9) Assemble the companion flange component to the drive pinion gear component.
(10) Apply transfer oil to the thread area of the new locknut.
(11) Assemble a washer and locknut to the drive pinion gear component.

cardiagn.com

Note
- If the locknut cannot be assembled, assemble the companion flange by lightly tapping it with a plastic hammer to acquire threading on the drive pinion gear.

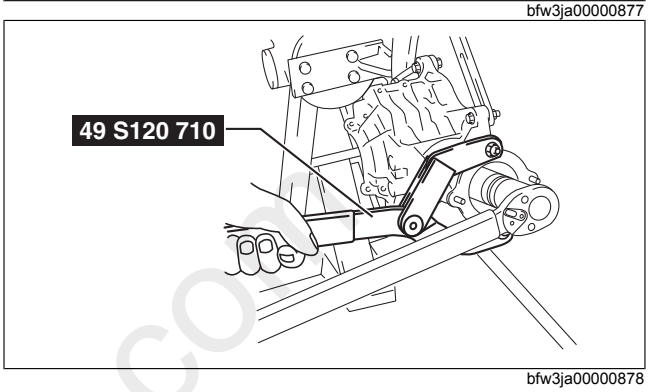
(12)Rotate the companion flange by hand to seat the bearing.


bfw3ja00000877

(13)Using the **SST**, gradually tighten the locknut from the minimum specified tightening torque and adjust it so that the specified preload is obtained. Record the tightening torque at this time.

Tightening torque
80—284 N·m {8.2—28 kgf·m, 60—209 ft·lbf}

Drive pinion preload (With tapered roller bearing)
1.2—1.4 N·m {13—14 kgf·cm, 11—12 in·lbf}
Drive pinion preload (With ball bearing/tandem ball bearing)
0.3—0.6 N·m {3.1—6.1 kgf·cm, 2.7—5.3 in·lbf}



bfw3ja00000878

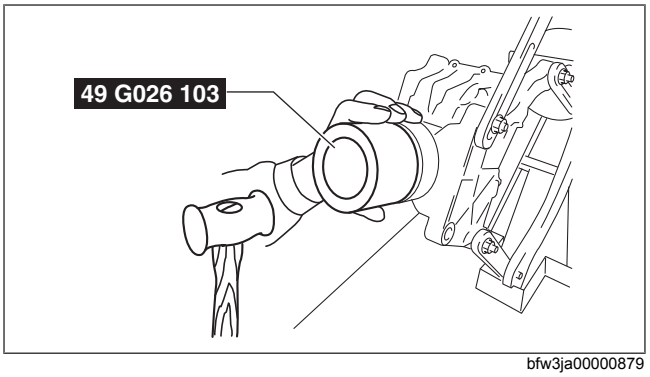
- If the specified preload is not obtained within the specified tightening torque, replace the collapsible spacer with a new one and perform the preload adjustment again.

(14)Remove the following parts from the front carrier. (See TRANSFER DISASSEMBLY [FW6AX-EL].)
- Locknut
- Washer
- Companion flange component

(15)Assemble a new oil seal using the **SST**.

Note
- Point the **SST** B-mark toward the oil seal side.

(16)Apply grease to the area of the companion flange where it contacts the bearing inner race (rear).
(17)Assemble the companion flange component to the drive pinion gear component.
(18)Apply transfer oil to the thread area of the new locknut.

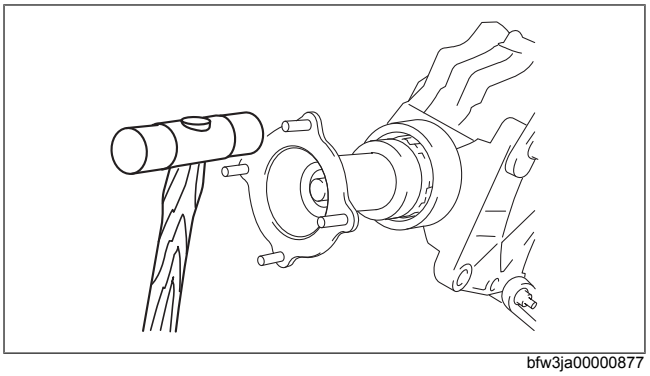


bfw3ja00000879

(19)Assemble a washer and locknut to the drive pinion gear component.

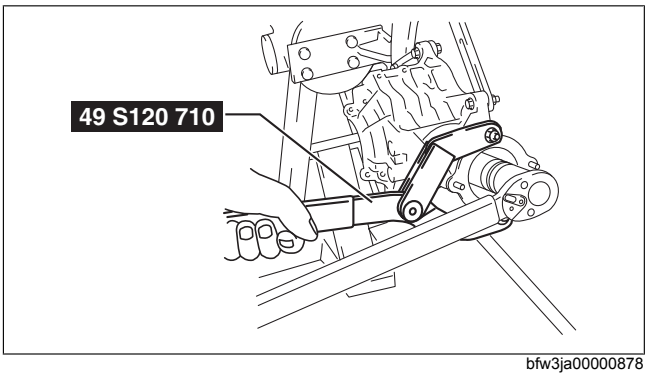
Note
- If the locknut cannot be assembled, assemble the companion flange by lightly tapping it with a plastic hammer to acquire threading on the drive pinion gear.

(20)Rotate the companion flange by hand to seat the bearing.


bfw3ja00000877

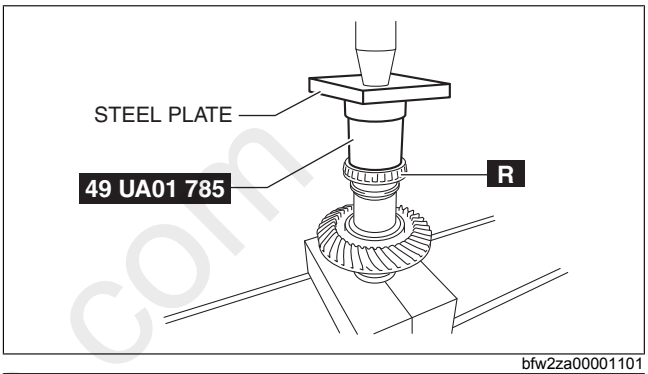
cardiagn.com

(21)Tighten the locknut to the torque recorded in drive pinion preload adjustment using the **SST**.



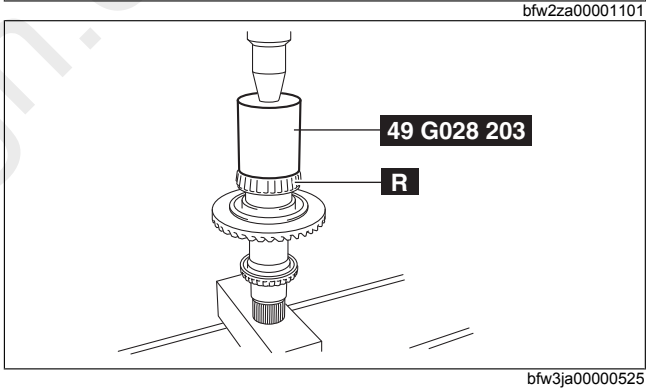
bfw3ja00000878

3. Assemble the ring gear shaft component to the front carrier using the following procedure.
   (1) Assemble the bearing inner race and bearing outer race to the ring gear shaft using the following procedure. (With tapered roller bearing)
      1) Press fit the bearing inner race (LH) to the ring gear shaft using the **SST** and a press, and a steel plate.



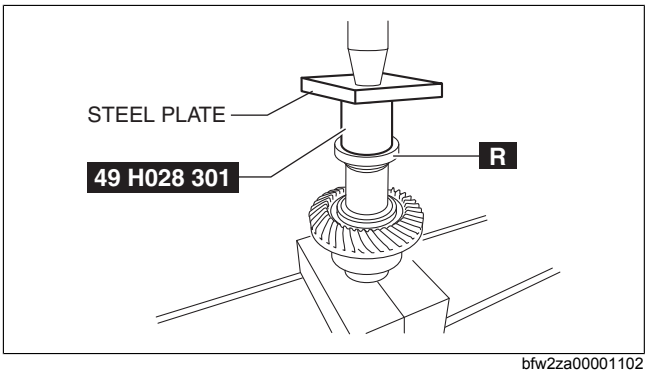
bfw2za00001101

      2) Press fit the bearing inner race (RH) to the ring gear shaft using the **SST** and a press.
      3) Temporarily assemble the bearing outer race (LH) and bearing outer race (RH) to the ring gear shaft.



bfw3ja00000525

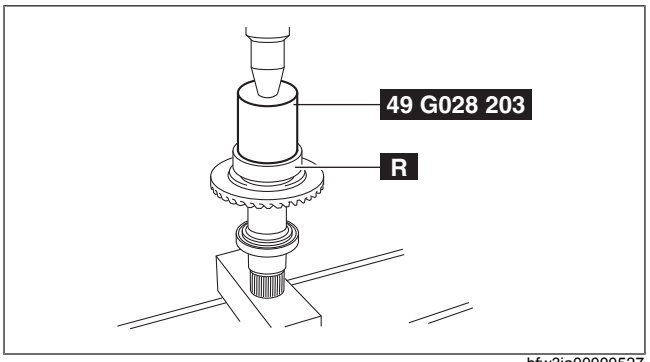
   (2) Assemble the ball bearing to the ring gear shaft using the following procedure. (With ball bearings/tandem ball bearings)
      1) Press fit the ball bearing (LH) to the ring gear shaft using the **SST** and a press, and a steel plate.



bfw2za00001102

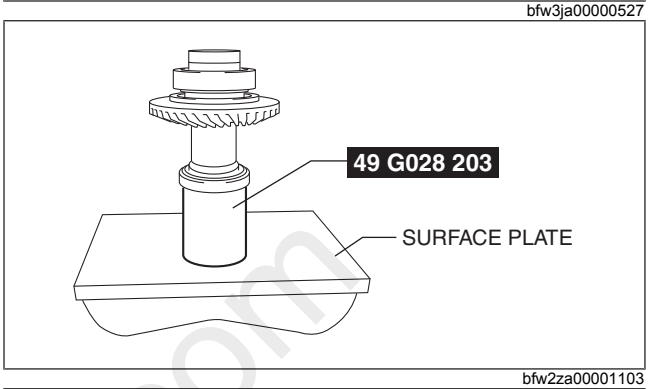
cardiagn.com

2) Press fit the bearing inner race (RH) to the ring gear shaft using the **SST** and a press.



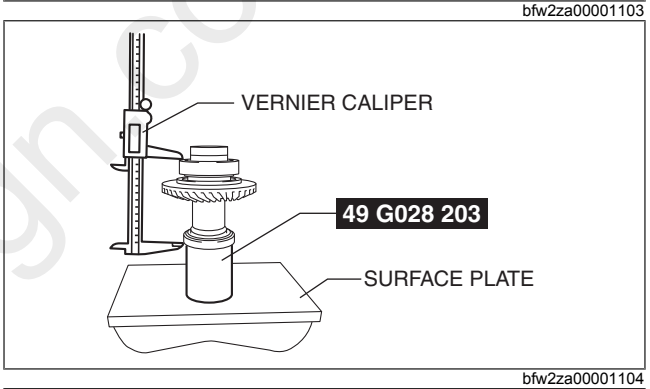
bfw3ja00000527

(3) Place the **SST** on a surface plate as shown in the figure and set the ring gear shaft component on top of it with the bearing outer race (LH) side pointed downward.
(4) Rotate the bearing by hand several times to seat it.



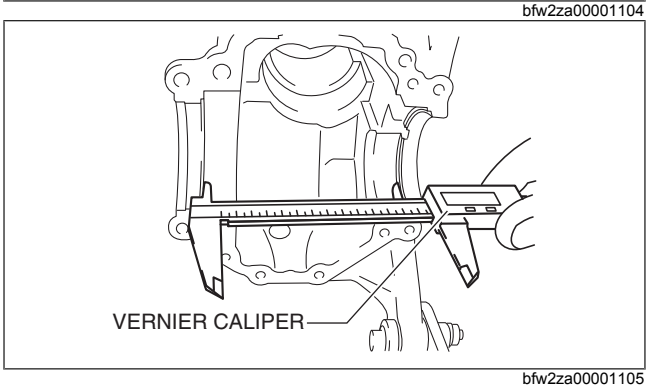
bfw2za00001103

(5) Using a vernier caliper, measure the distance between the bearing outer race (LH) and bearing outer race (RH).
- Measure any 3 locations and designate the average value as distance A.



bfw2za00001104

(6) Using a vernier caliper, measure the distance shown in the figure for the front carrier.
- Measure any 3 locations and designate the average value as distance B.

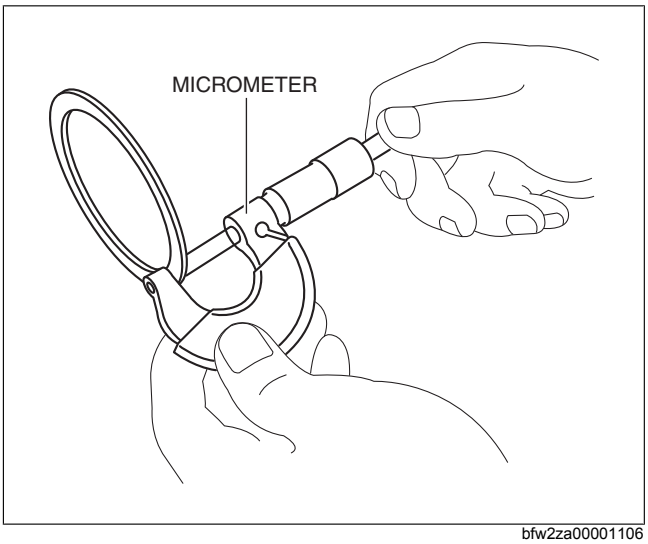


bfw2za00001105

carsdiagn.com

(7) Measure the thickness of adjustment shim (LH) and adjustment shim (RH) using a micrometer.
(8) Verify that the total value (thickness) of adjustment shim (LH) and adjustment shim (RH) is within the range of the minimum and maximum values.

With tapered roller bearing
Minimum value = Distance B - Distance A + 0.07 mm {0.003 in}
Maximum value = Distance B - Distance A + 0.16 mm {0.0063 in}
With ball bearings/tandem ball bearings
Minimum value = Distance B - Distance A + 0.1 mm {0.004 in}
Maximum value = Distance B - Distance A + 0.17 mm {0.0067 in}



bfw2za00001106

- If the total value (thickness) of adjustment shim (LH) and adjustment shim (RH) is within the minimum and maximum values, use them as is.
- If the total thickness of adjustment shim (LH) and adjustment shim (RH) is not within the range of the maximum and minimum values, select the appropriate adjustment shims from the following table.

Adjustment shim table (With tapered roller bearing)

Identification number	Thickness (mm {in})	Identification number	Thickness (mm {in})
449	4.490 {0.1768}	503	5.030 {0.1980}
452	4.520 {0.1780}	506	5.060 {0.1992}
455	4.550 {0.1791}	509	5.090 {0.2004}
458	4.580 {0.1803}	512	5.120 {0.2016}
461	4.610 {0.1815}	515	5.150 {0.2028}
464	4.640 {0.1827}	518	5.180 {0.2039}
467	4.670 {0.1839}	521	5.210 {0.2051}
470	4.700 {0.1850}	524	5.240 {0.2063}
473	4.730 {0.1862}	527	5.270 {0.2075}
476	4.760 {0.1874}	530	5.300 {0.2087}
479	4.790 {0.1886}	533	5.330 {0.2098}
482	4.820 {0.1898}	536	5.360 {0.2110}
485	4.850 {0.1909}	539	5.390 {0.2122}
488	4.880 {0.1921}	542	5.420 {0.2134}
491	4.910 {0.1933}	545	5.450 {0.2146}
494	4.940 {0.1945}	548	5.480 {0.2157}
497	4.970 {0.1957}	551	5.510 {0.2169}
500	5.000 {0.1969}	–	–

Adjustment shim table (With ball bearing/tandem ball bearing)

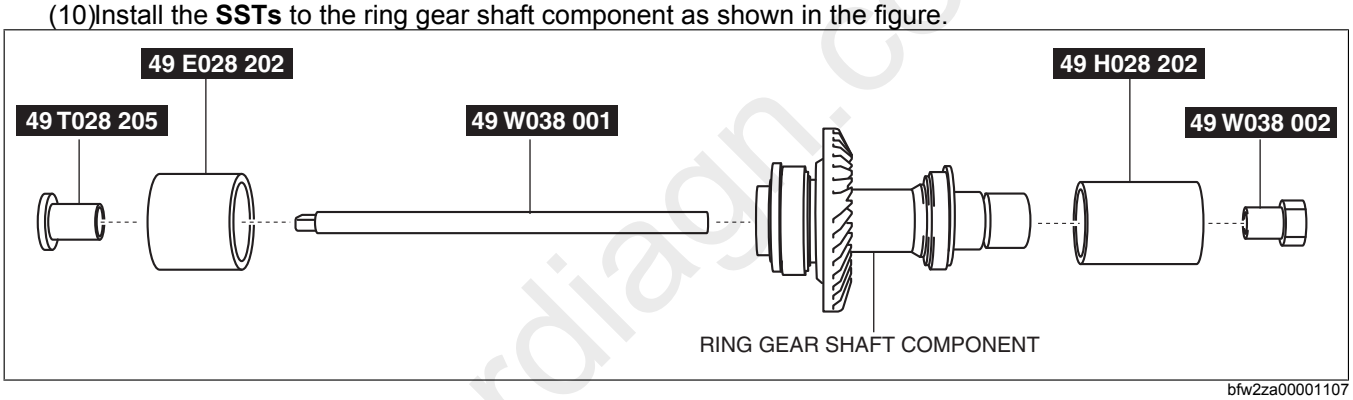
Identification number	Thickness (mm {in})	Identification number	Thickness (mm {in})
6490	6.490 {0.2555}	7000	7.000 {0.2756}
6505	6.505 {0.2561}	7015	7.015 {0.2762}
6520	6.520 {0.2567}	7030	7.030 {0.2768}
6535	6.535 {0.2573}	7045	7.045 {0.2774}
6550	6.550 {0.2579}	7060	7.060 {0.2780}
6565	6.565 {0.2585}	7075	7.075 {0.2785}
6580	6.580 {0.2591}	7090	7.090 {0.2791}
6595	6.595 {0.2596}	7105	7.105 {0.2797}
6610	6.610 {0.2602}	7120	7.120 {0.2803}
6625	6.625 {0.2608}	7135	7.135 {0.2809}
6640	6.640 {0.2614}	7150	7.150 {0.2815}
6655	6.655 {0.2620}	7165	7.165 {0.2821}
6670	6.670 {0.2626}	7180	7.180 {0.2827}
6685	6.685 {0.2632}	7195	7.195 {0.2833}
6700	6.700 {0.2638}	7210	7.210 {0.2839}
6715	6.715 {0.2644}	7225	7.225 {0.2844}
6730	6.730 {0.2650}	7240	7.240 {0.2850}
6745	6.745 {0.2656}	7255	7.255 {0.2856}

carmagn.com

Identification number	Thickness (mm {in})	Identification number	Thickness (mm {in})
6760	6.760 {0.2661}	7270	7.270 {0.2862}
6775	6.775 {0.2667}	7285	7.285 {0.2868}
6790	6.790 {0.2673}	7300	7.300 {0.2874}
6805	6.805 {0.2679}	7315	7.315 {0.2880}
6820	6.820 {0.2685}	7330	7.330 {0.2886}
6835	6.835 {0.2691}	7345	7.345 {0.2892}
6850	6.850 {0.2697}	7360	7.360 {0.2898}
6865	6.865 {0.2703}	7375	7.375 {0.2904}
6880	6.880 {0.2709}	7390	7.390 {0.2909}
6895	6.895 {0.2715}	7405	7.405 {0.2915}
6910	6.910 {0.2720}	7420	7.420 {0.2921}
6925	6.925 {0.2726}	7435	7.435 {0.2927}
6940	6.940 {0.2732}	7450	7.450 {0.2933}
6955	6.955 {0.2738}	7465	7.465 {0.2939}
6970	6.970 {0.2744}	7480	7.480 {0.2945}
6985	6.985 {0.2750}	7495	7.495 {0.2951}
–	–	7510	7.510 {0.2957}

(9) Assemble the following parts to the ring gear shaft.
- Bearing outer race (LH) (With tapered roller bearing)
- Bearing outer race (RH) (With tapered roller bearing)
- Adjustment shim (LH)
- Adjustment shim (RH)

(10) Install the **SSTs** to the ring gear shaft component as shown in the figure.

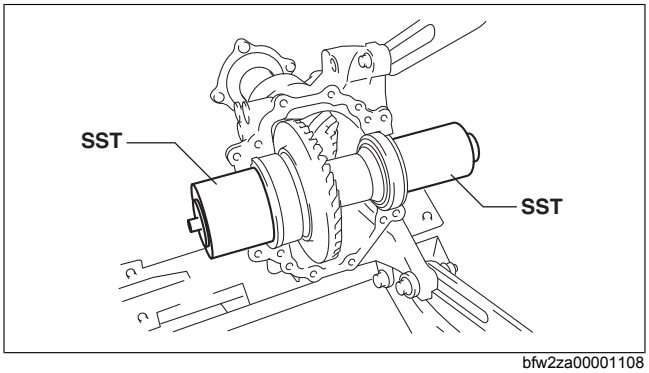


bfw2za00001107

(11) Tighten down the bolts on the left and right of the **SST** until the ring gear shaft component is assembled to the front carrier.

(12) Assemble the ring gear shaft component with the **SSTs** to the front carrier.

(13) Remove the **SSTs** from the ring gear shaft component.



bfw2za00001108

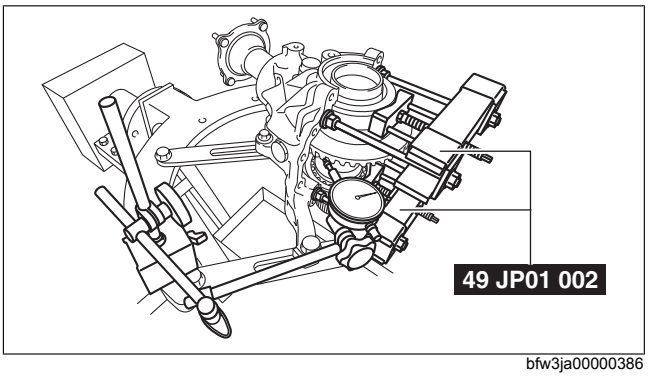
4. Perform the ring gear and drive pinion backlash adjustment.

Note
- Assemble the **SST** so that it does not contact the ring gear.

cardiagn.com

(1) Assemble the **SSTs** and dial gauge to the differential carrier as shown in the figure.

Tightening torque
5.0 N·m {51 kgf·cm, 44 in·lbf}

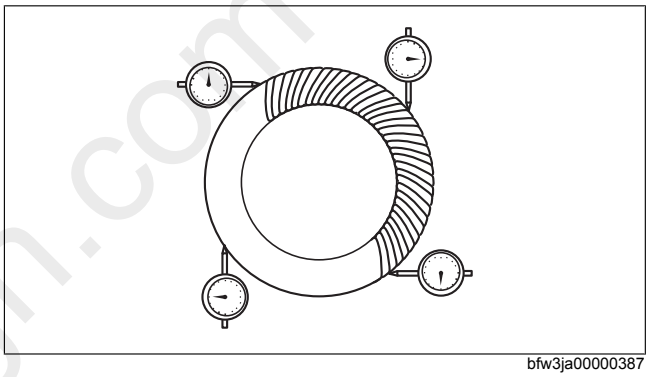


bfw3ja00000386

(2) Secure the drive pinion gear and measure the backlash from when the ring gear is moved.

Drive pinion gear and ring gear backlash
Standard: 0.09—0.14 mm {0.004—0.005 in}
Minimum: 0.05 mm {0.002 in}
Variance: 0.07 mm {0.003 in}

Note
- Measure the backlash at 4 locations around the ring gear. Verify that 1 of the 4 locations is within the standard, and the minimum value for the 4 locations is **0.05 mm {0.002 in} or more**, and the variance is **0.07 mm {0.003 in} or less**.
  - If the measured value of the backlash is not within the standard, adjust it by moving the ring gear shaft in the axial direction.


bfw3ja00000387

Note
- Replace the adjustment shims to move the ring gear shaft in the axial direction. At this time, if the adjustment shim on the right side is replaced with one 0.03 mm thicker, replace the adjustment shim on the left side with one **0.03 mm** thinner.

Adjustment shim table (With tapered roller bearing)

Identification number	Thickness (mm {in})	Identification number	Thickness (mm {in})
449	4.490 {0.1768}	503	5.030 {0.1980}
452	4.520 {0.1780}	506	5.060 {0.1992}
455	4.550 {0.1791}	509	5.090 {0.2004}
458	4.580 {0.1803}	512	5.120 {0.2016}
461	4.610 {0.1815}	515	5.150 {0.2028}
464	4.640 {0.1827}	518	5.180 {0.2039}
467	4.670 {0.1839}	521	5.210 {0.2051}
470	4.700 {0.1850}	524	5.240 {0.2063}
473	4.730 {0.1862}	527	5.270 {0.2075}
476	4.760 {0.1874}	530	5.300 {0.2087}
479	4.790 {0.1886}	533	5.330 {0.2098}
482	4.820 {0.1898}	536	5.360 {0.2110}
485	4.850 {0.1909}	539	5.390 {0.2122}
488	4.880 {0.1921}	542	5.420 {0.2134}
491	4.910 {0.1933}	545	5.450 {0.2146}
494	4.940 {0.1945}	548	5.480 {0.2157}
497	4.970 {0.1957}	551	5.510 {0.2169}
500	5.000 {0.1969}	–	–

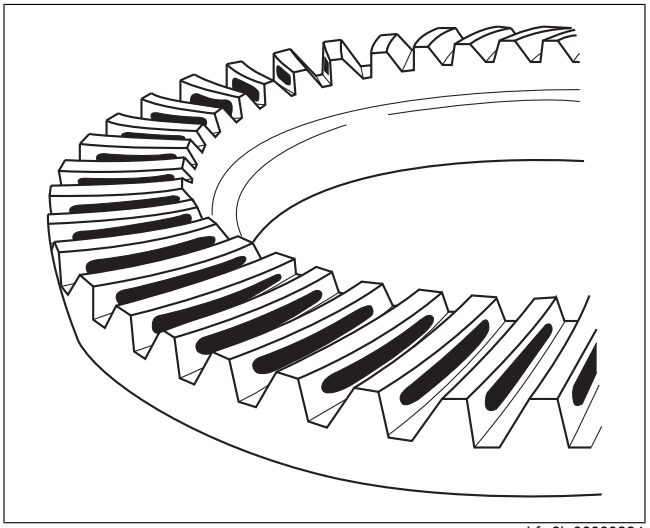
Adjustment shim table (With ball bearing/tandem ball bearing)

Identification number	Thickness (mm {in})	Identification number	Thickness (mm {in})
6490	6.490 {0.2555}	7000	7.000 {0.2756}
6505	6.505 {0.2561}	7015	7.015 {0.2762}
6520	6.520 {0.2567}	7030	7.030 {0.2768}
6535	6.535 {0.2573}	7045	7.045 {0.2774}

carmanuals.com

Identification number	Thickness (mm {in})	Identification number	Thickness (mm {in})
6550	6.550 {0.2579}	7060	7.060 {0.2780}
6565	6.565 {0.2585}	7075	7.075 {0.2785}
6580	6.580 {0.2591}	7090	7.090 {0.2791}
6595	6.595 {0.2596}	7105	7.105 {0.2797}
6610	6.610 {0.2602}	7120	7.120 {0.2803}
6625	6.625 {0.2608}	7135	7.135 {0.2809}
6640	6.640 {0.2614}	7150	7.150 {0.2815}
6655	6.655 {0.2620}	7165	7.165 {0.2821}
6670	6.670 {0.2626}	7180	7.180 {0.2827}
6685	6.685 {0.2632}	7195	7.195 {0.2833}
6700	6.700 {0.2638}	7210	7.210 {0.2839}
6715	6.715 {0.2644}	7225	7.225 {0.2844}
6730	6.730 {0.2650}	7240	7.240 {0.2850}
6745	6.745 {0.2656}	7255	7.255 {0.2856}
6760	6.760 {0.2661}	7270	7.270 {0.2862}
6775	6.775 {0.2667}	7285	7.285 {0.2868}
6790	6.790 {0.2673}	7300	7.300 {0.2874}
6805	6.805 {0.2679}	7315	7.315 {0.2880}
6820	6.820 {0.2685}	7330	7.330 {0.2886}
6835	6.835 {0.2691}	7345	7.345 {0.2892}
6850	6.850 {0.2697}	7360	7.360 {0.2898}
6865	6.865 {0.2703}	7375	7.375 {0.2904}
6880	6.880 {0.2709}	7390	7.390 {0.2909}
6895	6.895 {0.2715}	7405	7.405 {0.2915}
6910	6.910 {0.2720}	7420	7.420 {0.2921}
6925	6.925 {0.2726}	7435	7.435 {0.2927}
6940	6.940 {0.2732}	7450	7.450 {0.2933}
6955	6.955 {0.2738}	7465	7.465 {0.2939}
6970	6.970 {0.2744}	7480	7.480 {0.2945}
6985	6.985 {0.2750}	7495	7.495 {0.2951}
–	–	7510	7.510 {0.2957}

5. Perform the drive pinion gear and ring gear shaft tooth contact inspection using the following procedure.
   (1) Apply tooth marking compound (red lead coating) evenly to both tooth surfaces of the ring gear.
   (2) Rotate the ring gear back and forth several times.
   (3) Inspect the tooth contact pattern in 4 locations around the ring gear and verify that the tooth contact points exhibit the pattern shown in the figure.

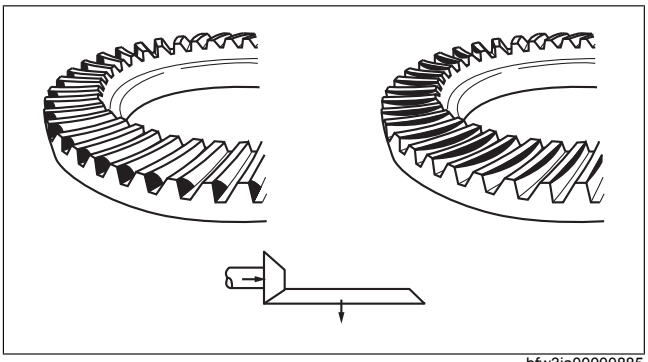


bfw3ja00000884

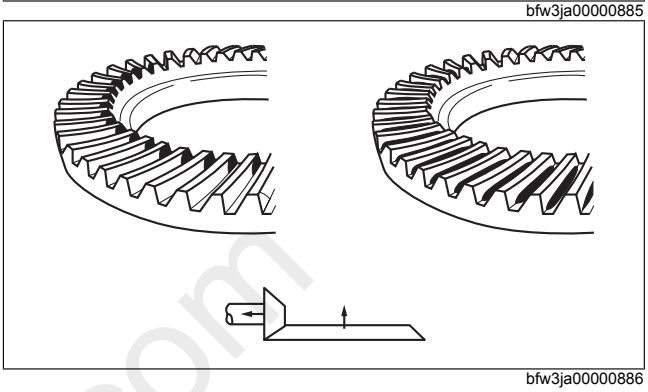
- If the tooth contact points are normal, wipe off the marking compound.
- If the tooth contact is not correct, adjust the drive pinion gear spacer. Then, adjust the drive pinion preload and the backlash.

carfiagn.com

— If the heel and face contact points appear as shown in the figure, replace the drive pinion gear spacer with a thicker one to move the drive pinion gear closer to the ring gear.


bfw3ja00000885

— If the tow and flank contact points appear as shown in the figure, replace the drive pinion gear spacer with a thinner one to move the drive pinion gear farther away from the ring gear.


bfw3ja00000886

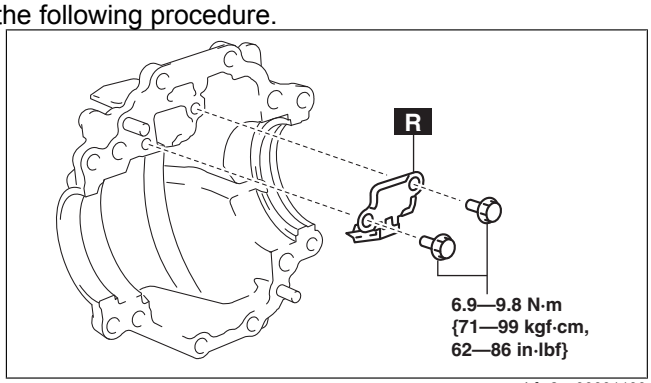
Spacer table

Identification number	Thickness (mm {in})	Identification number	Thickness (mm {in})
08	3.080 {0.1213}	29	3.290 {0.1295}
09	3.095 {0.1219}	30	3.305 {0.1301}
11	3.110 {0.1224}	32	3.320 {0.1307}
12	3.125 {0.1230}	33	3.335 {0.1313}
14	3.140 {0.1236}	35	3.350 {0.1319}
15	3.155 {0.1242}	36	3.365 {0.1325}
17	3.170 {0.1248}	38	3.380 {0.1331}
18	3.185 {0.1254}	39	3.395 {0.1337}
20	3.200 {0.1260}	41	3.410 {0.1343}
21	3.215 {0.1266}	42	3.425 {0.1348}
23	3.230 {0.1272}	44	3.440 {0.1354}
24	3.245 {0.1278}	45	3.455 {0.1360}
26	3.260 {0.1283}	47	3.470 {0.1366}
27	3.275 {0.1289}	–	–

6. Assemble the drive gear case to the front carrier using the following procedure.
   (1) Assemble a new baffle plate to the drive gear case.

   Tightening torque
   6.9—9.8 N·m {71—99 kgf·cm, 62—86 in·lbf}

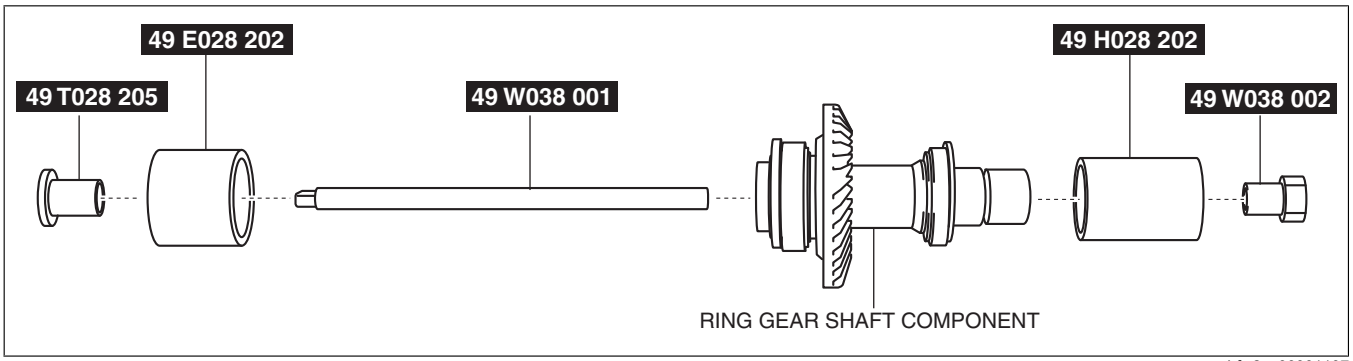
   (2) Verify that there is no gap between the baffle plate seal area lip and the drive gear case.



bfw2za00001109

   (3) Install the **SSTs** to the ring gear shaft component as shown in the figure.

carmagn.com

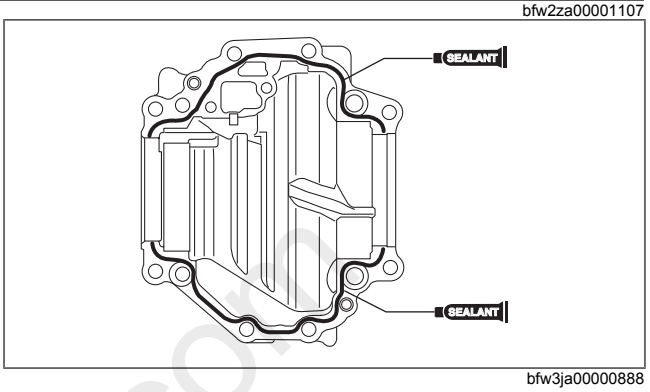


bfw2za00001107

(4) Apply a thin layer of sealant (silicone sealant TB1217C or equivalent) to the locations indicated in the figure for the drive gear case.

Caution

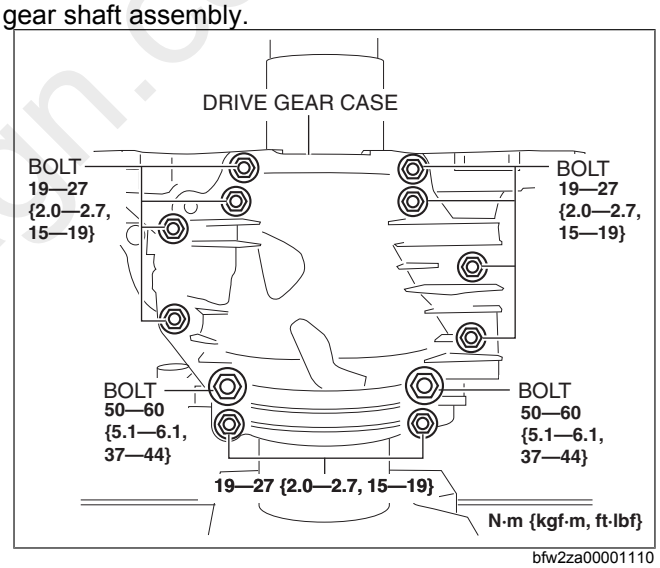
- **To prevent the silicone sealant from hardening, adhere the drive gear case to the front carrier within 3 min after applying silicone sealant.**



bfw3ja00000888

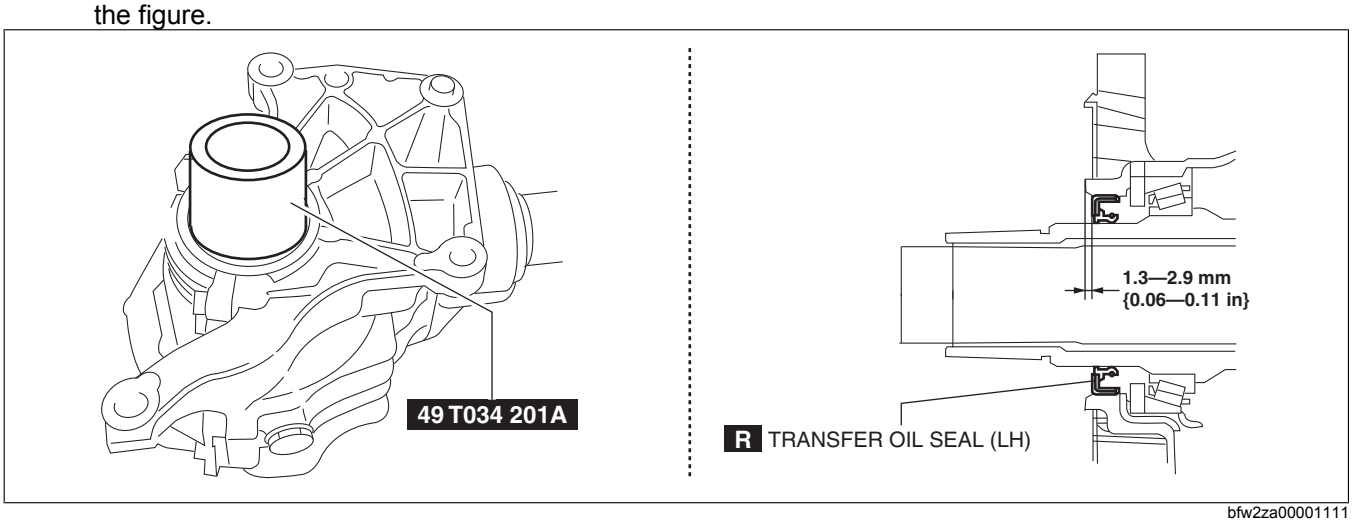
(5) Tighten the bolts on the left and right of the **SSTs** assembled to the ring gear shaft component, and assemble the drive gear case to the front carrier with pressure applied to the ring gear shaft assembly.

(6) Tighten the drive gear case assembly bolts to the specified torque.



bfw2za00001110

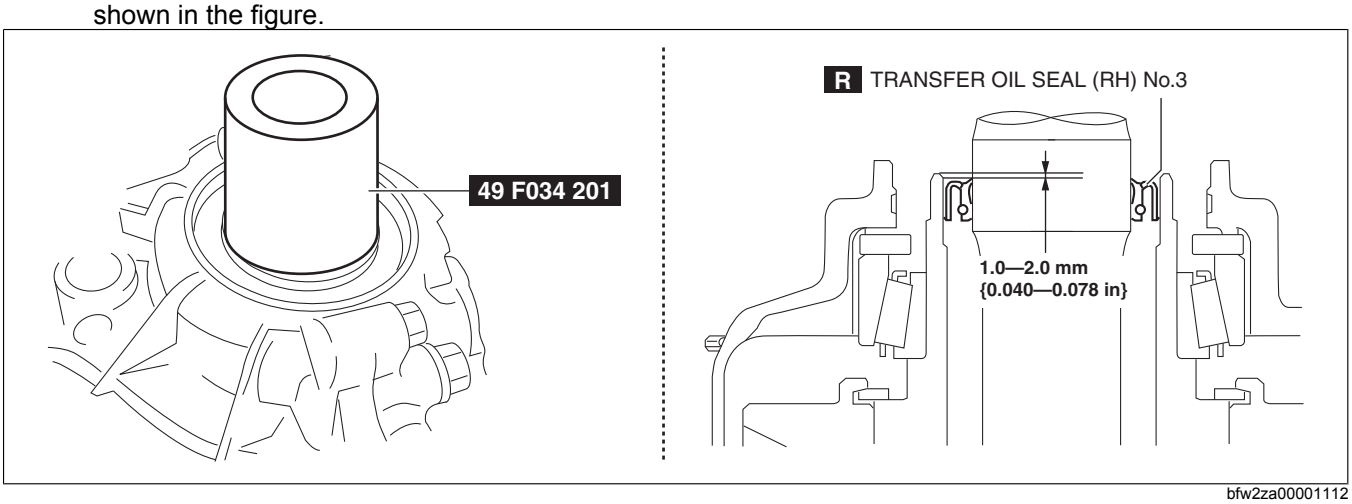
7. Assemble the transfer oil seal to the transfer using the following procedure.
   (1) Using the **SST** and a hammer, assemble a new transfer oil seal (LH) according to the dimensions shown in the figure.



bfw2za00001111

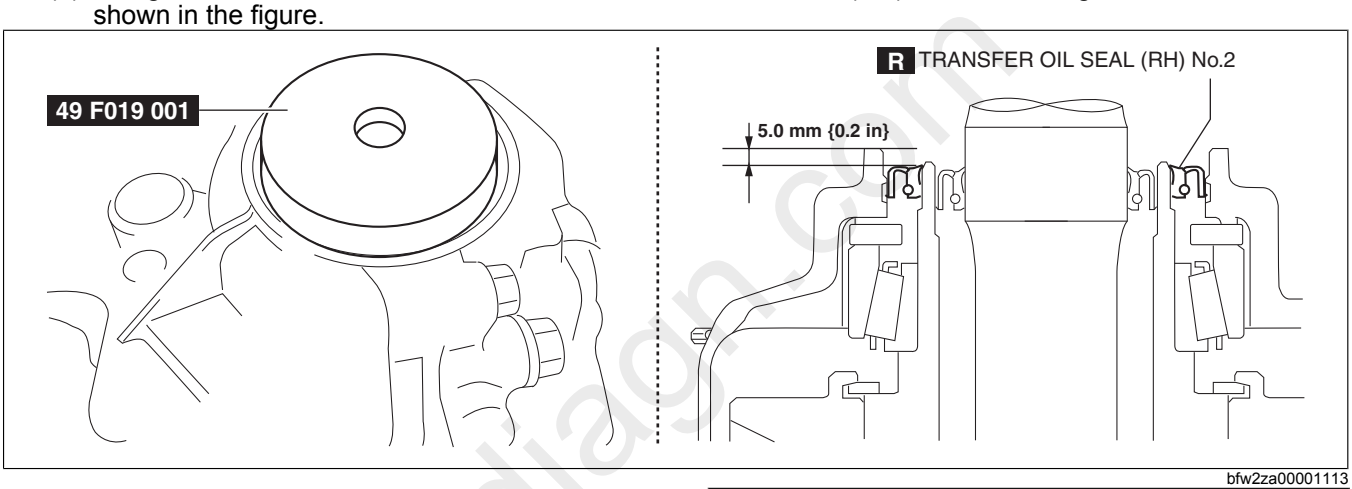
cardiagn.com

(2) Using the **SST** and a hammer, assemble a new transfer oil seal (RH) No.3 according to the dimensions shown in the figure.



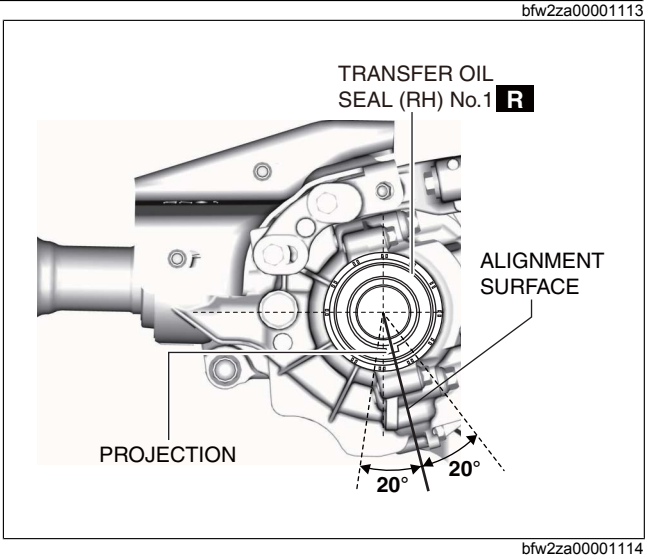
bfw2za00001112

(3) Using the **SST** and a hammer, assemble a new transfer oil seal (RH) No.2 according to the dimensions shown in the figure.



bfw2za00001113

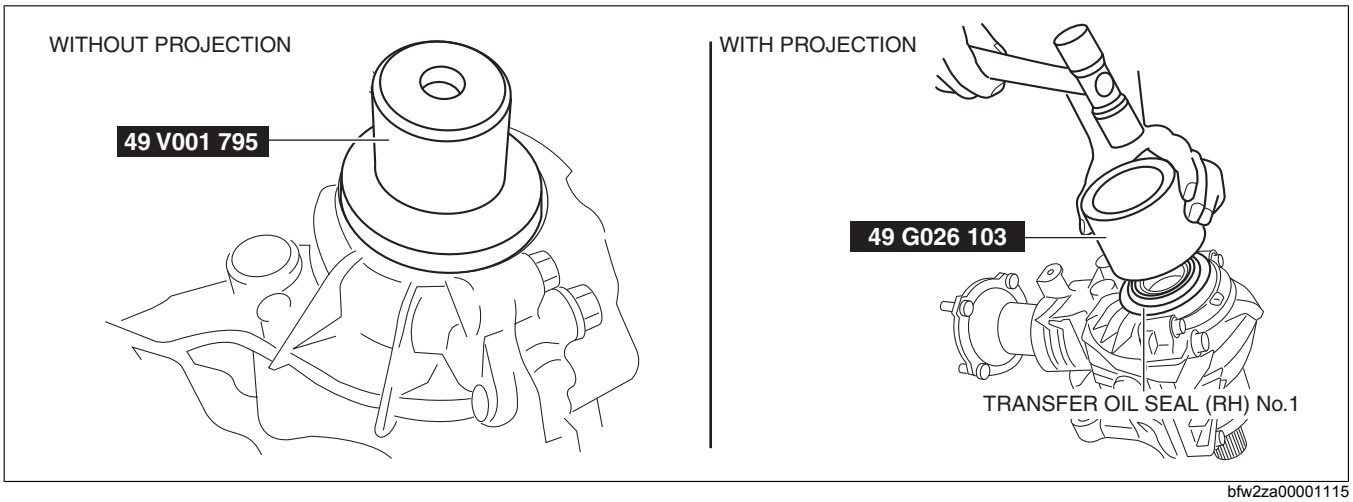
(4) Assemble the transfer oil seal (RH) No.1 so that the projection of the transfer oil seal (RH) No.1 is within the range shown in the figure. (With projection)



bfw2za00001114

(5) Using the **SST** and a hammer, assemble a new transfer oil seal (RH) No.1.

carlagn.com



bfw2za00001115

8. Assemble a new breather.

   Tightening torque
   8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}

9. Assemble a new washer and drain plug.

   Tightening torque
   40—58 N·m {4.1—5.9 kgf·m, 30—42 ft·lbf}

10. Assemble a new washer and oil level plug.

    Tightening torque
    40—58 N·m {4.1—5.9 kgf·m, 30—42 ft·lbf}

cardiagn.com